Enhanced Natural Language Interaction Platform With Offensive Content Filtering

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Abstract

- The rapid growth of online messaging has introduced a significant rise in hateful speech and abusive language.
- This system is a real-time chat application integrating Natural Language Processing (NLP) to detect and filter offensive content.
- Built on the MERN stack, it uses deep learning to analyze, block, or replace inappropriate messages before delivery , promoting ethical AI usage.

Objective

- To develop a real-time, professional chat application using the MERN stack and Socket.IO.
- To detect and filter inappropriate or offensive messages pre-delivery.
- To implement a hybrid deep learning architecture (DistilBERT + GRU) for high-accuracy toxicity detection.
- To replace slang and abusive words with contextually appropriate alternatives.

Dataset Details

MetaHate
Source: IRLab-UDC on huggingface.co
Total Test Samples: 165,175
Class Distribution: * ~79% Non-Toxic | ~21% Toxic
Model Parameters:
128 Max Tokens | 50,000 Vocab

System Design

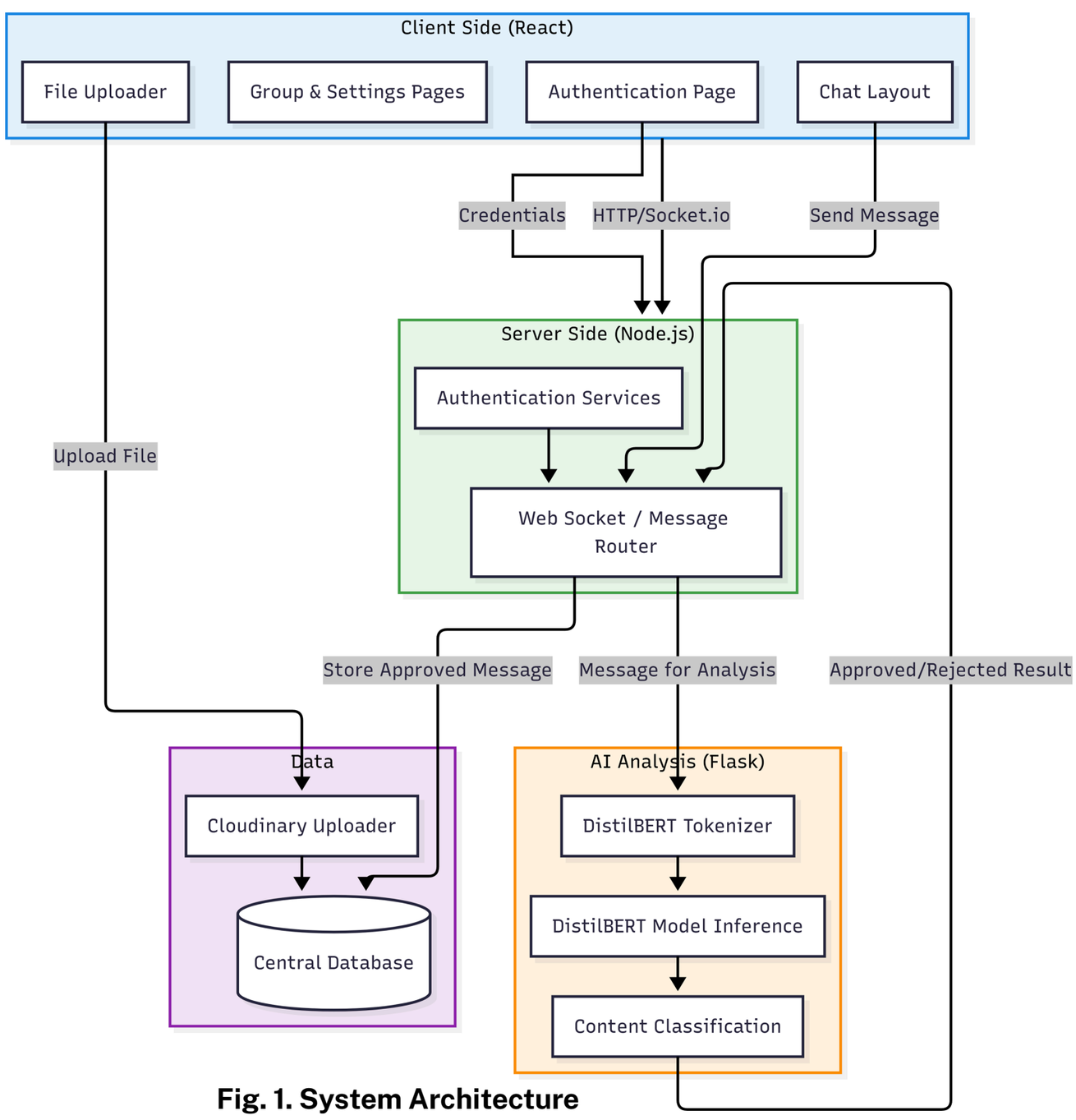

Fig. 1. System Architecture

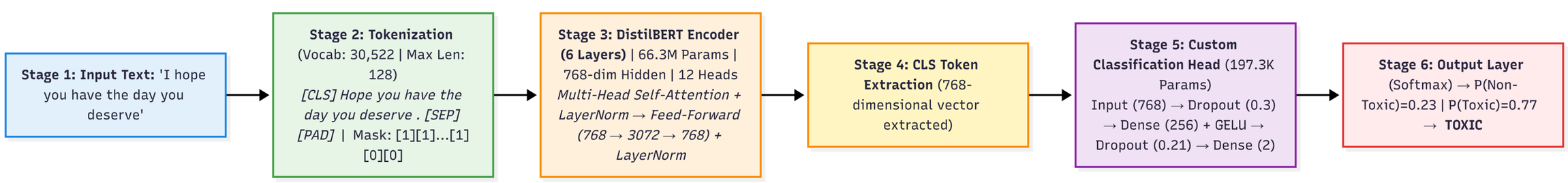

Fig. 2. Model Architecture

python node JS React JS TensorFlow mongoDB

Fig. 3. Tools and Technologies

Model Results

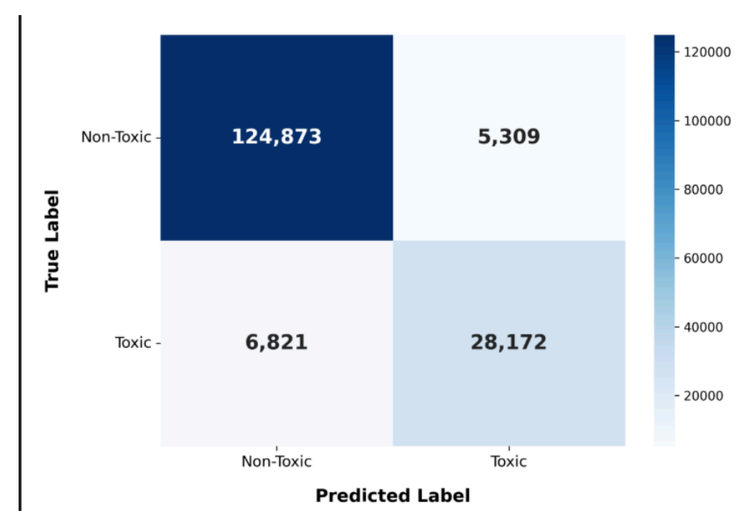

Fig. 4. Confusion Matrix

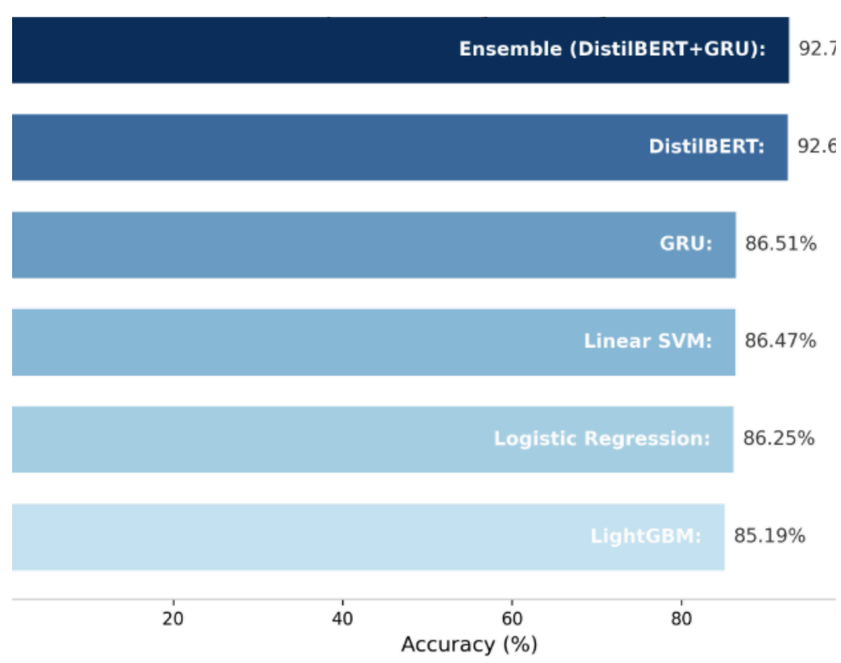

Fig. 5. Top Models Comparison

SDG Goal

Conclusion

- This project demonstrates the effective deployment of an ethical communication platform by combining modern web technologies (MERN, Socket.IO) with advanced NLP.
- The integration of a deep learning ensemble yields a highly accurate (>92%) toxicity classification , successfully automating the detection, blocking, and replacement of abusive content.

References

1. "Addressing challenges in hate speech detection using BERT-based models: A review," ResearchGate, 2024
2. "Deep learning-based approaches for abusive content detection and classification," ScienceDirect, 2024.
3. "Deep learning and natural language processing in computation for offensive language detection," Theoretical Computer Science, vol. 943, pp. 203–216, Jan. 2023.